Recoverable Error in Neural Climate Compression: A Rate-Matched Study of CRA5 over CONUS

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Abstract. Learned atmospheric compressors are usually assessed by aggregate distortion, which does not reveal how much residual error remains predictable from decoded fields. We apply a fixed 1.65 M-parameter U-Net to CRA5 reconstructions over CONUS without adding per-frame bits. On the full 2018 test year, mean normalised RMSE decreases by 25.8% over 19 directly represented FourCastNet variables (95% temporal block-bootstrap interval [25.32, 26.18]%). On the complete 2022 year outside CRA5's published training window, the same frozen checkpoint and normalisation give a 26.4% reduction [25.90, 26.93]%, without fine-tuning. Near-surface versus upper-air gains are 32.2% versus 3.5% in 2018 and 32.8% versus 3.6% in 2022. A mean-rate-matched MPCA-only control has a smaller near/upper gain ratio, while additional MPCA+PCA controls are explicitly non-matched. A parameter-matched dilation ablation retains a roughly eightfold split and concentrated gradient sensitivity. Using ERA5 orography and land–sea masks, western-land correction gains increase from 18.1% to 36.9% between the lowest and highest terrain-roughness deciles, while corrected errors remain larger on rough ground. Elevation-stratified comparisons support an association but do not isolate a terrain mechanism. These results measure model-relative predictability; terrain causality, corrector architecture effects, and broader distribution-shift robustness remain open. Shared weights and inference add deployment costs.

Keywords: Scientific data compression · Atmospheric reanalysis · Learned compression · Decoder-side residual prediction · ERA5

1 Introduction

ERA5 [7] exceeds 400 TB in float32, making storage and data movement practical constraints. CRA5 [5,6] uses a transformer variational autoencoder to reduce this archive to about 0.85 TB. Its published evaluation describes reconstruction error by variable, pressure level, spatial location, and spectrum. A separate question is how much error remains predictable from the decoded fields themselves.

We attach a 1.65 M-parameter U-Net to CRA5's decoded output. On the 2018 CONUS test year, mean normalised RMSE (nRMSE) decreases by 25.8%

over the 19 FourCastNet variables directly represented by CRA5, without changing the per-frame bitstream. Including derived total column water vapour gives 27.8%. We call this reduction *model-recoverable error*: it depends on the chosen network and training protocol and is not an information-theoretic bound. On the complete 2022 year outside CRA5's published training window, the same frozen corrector reduces direct-channel nRMSE by 26.4% without fine-tuning.

The reduction is uneven: 32.2% for near-surface fields versus 3.5% aloft. Their 9.1-fold ratio compares correction gains, not compression ratios. We apply the same correction architecture, with separately fitted weights, to classical transform codecs. A single-stage MPCA-only codec matched to CRA5's mean bits per value gives a near/upper gain ratio of 1.7; three MPCA+PCA settings at higher compression give 1.32–1.35. These controls help assess whether the asymmetry is specific to the tested codec and corrector combination.

Terrain is a plausible contributor: small-scale topography may be difficult to represent on a coarse latent grid, but error magnitude and correction gain are different quantities. Using actual ERA5 orography and land–sea masks, we find larger baseline errors and larger gains on rougher western land: G rises from 18.1% to 36.9% between roughness endpoint deciles. The association persists within several elevation bands but does not isolate a terrain mechanism. Our earlier proxy-elevation analysis remains a sensitivity check.

Our contributions are (i) a measured, mean-rate-matched MPCA/CRA5 comparison with the differing global/regional coding tasks made explicit; (ii) a study of decoder-side residual predictability and its vertical asymmetry, including a frozen 2022 out-of-training-window test and corrector controls; and (iii) an ERA5-orography analysis of terrain roughness versus correction gain, with elevation and geographic sensitivity checks. Checkpoints are selected using 2017 validation data; paired temporal block-bootstrap intervals quantify uncertainty conditional on each fitted checkpoint and test year.

2 Related Work

CRA5 [5,6] reports per-variable and per-level errors, climatological bias maps, zonal spectra, and surface-station comparisons. Its descriptor associates elevated error over mountain regions with topography. Mirowski et al. [11] provide elevation and a land–sea mask to a reanalysis encoder. These results motivate terrain diagnostics, but spatial association alone does not isolate terrain effects or decoder-side predictability. Ding et al. [4] evaluate CRA5 for vertically integrated tropospheric delay, a different target from the horizontal residual structure studied here.

Decoder post-processing has precedent. Madireddy et al. [10] correct JPEG-compressed scientific simulation images with convolutional networks; Jian et al. [9] correct classical error-bounded compressors using a 3D CNN. Zhu et al. [13] correct frozen learned compressors by storing a quantised residual, while NeurLZ [8] also adds per-block information. Our corrector stores no per-frame residual and has no pointwise error guarantee.

Alhumaidi et al. [1] use CRA5, GraphCast, and Aurora as predictors within an error-bounded pipeline sharing SZ3.1's backend. Their residual maps and entropy analysis measure the cost of coding the remaining error; our experiment measures how much a fixed model can predict from decoded fields. These are complementary questions. Transform, quantisation, and entropy coding are standard components of scientific codecs, including TTHRESH [2]; the survey in [3] provides broader context. Our focus is the distribution of decoder-side correction gains, rather than a general ranking of neural and classical compressors.

3 Experimental Setup

3.1 Data and Metric

ERA5 [7] is represented on a regular 0.25° latitude–longitude grid and cropped to 25–49°N and 130–70°W (97×241 cells). We use the 20 FourCastNet variables [12]: five surface fields; winds and geopotential at 1000/850 hPa; temperature and relative humidity at 850 hPa; wind, geopotential, temperature, and relative humidity at 500 hPa; geopotential at 50 hPa; and total column water vapour. Frames are six-hourly at 00/06/12/18 UTC. Training uses 2014–2016 (4,384 frames), validation 2017 (1,460), and evaluation 2018 (1,460). Architecture and codec parameters are selected using validation data; reported quantisation steps are fixed before 2018 evaluation. The full-year 2022 test uses all 1,460 timestamps, January 1 at 00 UTC through December 31 at 18 UTC. ERA5 truth combines 18 native ARCO-ERA5 fields with native NCAR relative humidity at 500/850 hPa; decoded CRA5 and truth timestamps, coordinates, and channel order are checked exactly. Source packing precision is cross-checked against 2018 samples. The original checkpoint and 2014–2016 statistics are frozen; neither test year is used for fine-tuning or selection.

We compute channel nRMSE as $\sqrt{\langle (x_j - \hat{x}_j)^2 \rangle}/\sigma_j$. The mean pools test times and cells; σ_j is the training standard deviation. We average these channel errors equally. Primary CRA5 means use 19 direct channels; the full 20-channel mean is labelled separately. CRA5 does not store total column water vapour (`tcwv`); we derive it by pressure integration of specific humidity, truncated at the surface. Its $-0.63\,\mathrm{kg\,m^{-2}}$ bias can be learned by the corrector, so its gains partly repair this derivation.

For channel errors b_j before and a_j after correction, the headline reduction is $G = 1 - \sum_j a_j / \sum_j b_j$. Equal-weight fractional gain is $\bar{G} = p^{-1} \sum_j (1 - a_j/b_j)$. The former weights fractional gains by baseline error; the latter treats channels equally. Terrain roughness uses G; the legacy proxy-elevation control instead uses G_{pool} (Section 7).

3.2 Correction Network

All correctors train with 20 input/output channels; direct19 is only an evaluation subset. U-Net depth is 2, base/bottleneck widths 64/128, with 3×3 convolutions

and 1,646,548 parameters. Input is $(\hat{x}-\mu)/\sigma$, target $(x-\hat{x})/\sigma$, and corrected field $\hat{x}+\sigma f(\cdot)$; μ, σ are fixed training statistics. CRA5 correction minimises normalised residual MSE using AdamW: learning rate 3×10^{-4}, weight decay 10^{-4}, gradient clipping 1.0, batch 8, mixed precision, seed 0, maximum 200 epochs, and early-stopping patience 12. The architecture comes from a 33-configuration MPCA sweep ranked on 2017, not a CRA5 architecture search. Classical checkpoints share the architecture/objective but not identical budgets: inspected MPCA-only and e995 logs allow 200 and 100 epochs. Each is reused without retraining here.

We decode the released CRA5 streams for 2014–2018 and extract CONUS. Baseline 20-channel nRMSE ranges from 0.08523 to 0.08752 across these years; they overlap CRA5 training (Section 8). New 2018 diagnostics replay the original checkpoint with float32 inference on the existing float16 normalised cache and float64 aggregation, reproducing original rounded results.

Temporal uncertainty. Squared-error sums and counts for each frame and channel support a paired circular moving-block bootstrap: 28-frame (seven-day) blocks wrap at the year boundary, with each draw truncated to 1,460 frames. In 1,000 draws (seed 20261026), before/after errors share sampled timestamps; channel RMSEs and G are recomputed. Percentile 95% intervals condition on checkpoint, year, and grid, excluding training-seed and spatial uncertainty. The same protocol is applied to 2022.

4 A Rate-Matched Classical Encoder

The classical control uses multilinear PCA independently for each channel. Separate latitude and longitude bases retain 99% energy per axis; a core C reconstructs as $LCR^{\top}$. The twenty channels together retain 10,193 coefficients per frame. MPCA-only quantises these directly. MPCA+PCA adds a global PCA: e995, e99, and e97 retain 99.5%, 99%, and 97% energy, yielding 2,011, 1,449, and 636 components, respectively.

A uniform step Δ quantises the coefficients; a range coder uses a zero-mean Gaussian model with training-estimated scale for each coefficient. Every configuration is checked by an exact integer round-trip. All compression ratios (CRs) use measured coded bytes, not coefficient counts. For example, 2,011 float32 coefficients imply 232×, whereas the measured e995 stream achieves 1333×. Larger CR means lower bits per source value, $R = 32/\mathrm{CR}$.

The reported MPCA rate includes the scale table amortised over 7,304 frames (5.58 bytes per frame for MPCA-only). Transform bases are fixed codec parameters, as CRA5 weights are, and excluded from the per-frame payload. MPCA-only's bases contain 153,682 floats; including their amortised 84 bytes per frame would change its CR to approximately 463×. MPCA+PCA additionally requires a fixed global basis. These shared costs matter for short archives.

CRA5's 2018 streams average 2,360,056 bytes for $721 \times 1440 \times 268$ float32 values: 471.60×, or 0.0679 bits per value. Matching this rate for our $20\times97\times241$

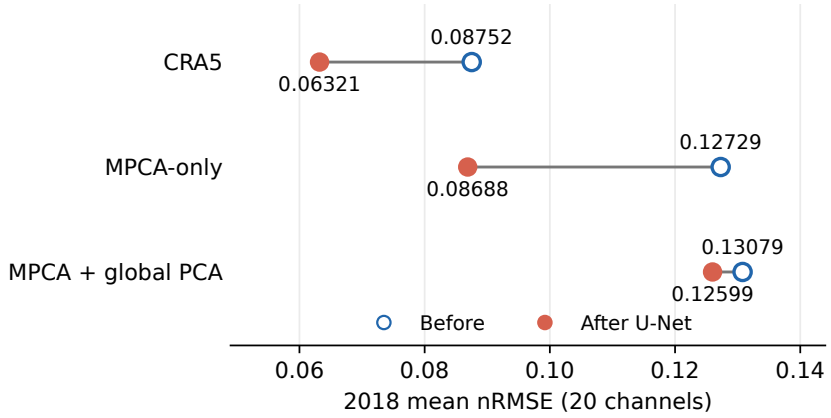


Fig. 1. Matched mean 2018 rates: CRA5 471.6×, MPCA-only 473.1×, and MPCA+global-PCA 472.9×. Hollow markers denote baseline and filled markers add correction without per-frame bits. Values average all 20 channels, including derived `tcwv`, matching the 20-channel columns in Table 1.

fields gives 3,965.6 bytes per frame. Bisection on 2017 coded sizes fixes $\Delta = 0.7659$; the frozen step yields 3,952.9 bytes and 473.12× on 2018, a 0.3% rate mismatch.

Figure 1's matched MPCA+PCA retains 4,384 components, distinct from e995/e99/e97. Matching concerns *mean bits per source value*. CRA5 jointly codes 268 global channels; MPCA is fitted to 20 regional channels. Their domains and coding tasks are different, and the global bitstream cannot identify CRA5's exact cost for this regional subset. The comparison is an operating-point control, not an equivalent-domain benchmark. MPCA-only is the main matched-rate classical control.

5 Results at Matched Rate

Figure 1 uses validation-selected quantisation; the 0.3% rate mismatch slightly favours CRA5. Correctors share an architecture/objective, with compressor-specific weights. Classical correctors train before quantisation and evaluate after it; CRA5 uses decoded streams throughout. Quantisation changes classical corrected nRMSE by 0.07–0.44%, but the mismatch remains a limitation.

CRA5 has lower error before and after correction (Fig. 1). Corrected MPCA-only (0.08688) is only 0.73% below uncorrected CRA5 (0.08752); different post-processing prevents a decisive compressor ranking. At matched rate, global PCA raises corrected error by 45% versus MPCA-only.

Decoder-side repair. CRA5 direct19 nRMSE falls from 0.08661 to 0.06431 ($G = 25.8\%$, $\bar{G} = 19.3\%$). With derived `tcwv`, $G = 27.8\%$.

Table 1. Full-year CRA5 correction, 1,460 frames/year. After$_{20}$ and G_{20} include derived `tcwv`, matching Fig. 1; baseline nRMSE is 0.08752/0.08700 in 2018/2022. All G columns are percentage reductions: G_{19} excludes `tcwv`; G_N, G_U use near13/upper6. All checkpoints are 2017-selected and frozen. Dilated is parameter-matched; Affine is not.

Corrector	Year	After$_{20}$	G_{20}	G_{19}	G_N	G_U
U-Net	2018	0.06321	27.78	25.75	32.18	3.52
	2022	0.06236	28.32	26.40	32.79	3.60
Dilated	2018	0.06354	27.40	25.38	31.61	3.82
	2022	0.06270	27.93	26.01	32.21	3.91
Affine	2018	0.08471	3.20	1.29	1.43	0.82
	2022	0.08425	3.16	1.31	1.45	0.85

The unchanged bitstream requires no residual side information; shared correction weights cost about 6.6 MB in float32 plus inference computation. This is model-relative predictability, not an information-theoretic bound. Profiling counts 19.86 G FLOPs per frame. On an RTX 3080, float32 GPU-resident inference at batch 8 averages 1.32 ms per frame with TF32 allowed; this excludes decoder, I/O, and transfer costs.

The 95% bootstrap intervals for direct19, near and upper G are [25.32, 26.18]%, [31.73, 32.62]% and [3.38, 3.68]%, respectively. The near-minus-upper difference is 28.66 percentage points [28.31, 29.01]. Table 2 gives group errors; intervals condition on checkpoint, year and grid.

Full-year out-of-time evaluation. Unlike 2018, the complete 2022 test year is outside CRA5's published training window. Table 1 shows that the frozen corrector retains its aggregate improvement: direct-channel $G = 26.40\%$ with a 95% block-bootstrap interval of [25.90, 26.93]%. Near-surface gain is 32.79% [32.28, 33.31]% versus 3.60% [3.46, 3.76]% aloft, a 9.10-fold ratio.

Including derived `tcwv`, 2022 nRMSE is 0.08700 before and 0.06236 after correction ($G = 28.32\%$); `tcwv` alone gains 59.43%. Aggregate gains do not imply uniform improvement: $v500$ worsens slightly (gain -0.0082%).

6 Where the Error Lives

Field altitude and variable identity. Near-surface means surface fields plus 1000/850 hPa (13 channels); upper-air means 500/50 hPa (6). These refer to field altitude, not ground elevation; `tcwv` spans the atmospheric column and belongs to neither group. The reductions are $G = 32.2\%$ versus 3.5% and $\bar{G} = 27.5\%$ versus 1.5%. Their ratios, 9.1 and 18.6, compare gains rather than compression. Surface-pressure nRMSE falls from 0.0990 to 0.0249; the 500 hPa wind errors are unchanged to four decimals. The largest upper-air gain is 7.4% for relative humidity at 500 hPa.

Table 2. Full-2018 frozen-checkpoint evaluation (1,460 frames). Before/after nRMSE averages 13 near-surface or 6 upper-air direct channels; both exclude column-integrated `tcwv`. G is their relative reduction. CR uses measured entropy-coded bytes: R is CRA5 reference, M is matched mean rate, and N is non-matched. e995/e99/e97 denote MPCA+PCA. Each system reuses its own previously trained weights; MPCA-only was trained on stage-1-only reconstruction and is not retrained for this table. Classical rows use actual quantised reconstruction at the original fixed steps.

System	CR/status	Group	Before	After	G (%)
CRA5	471.6 R	Near	0.09820	0.06660	32.18
		Upper	0.06151	0.05934	3.52
MPCA-only	473.1 M	Near	0.12913	0.08207	36.44
		Upper	0.12289	0.09690	21.14
e995	1333 N	Near	0.17983	0.14221	20.92
		Upper	0.15849	0.13392	15.50
e99	2000 N	Near	0.19156	0.15536	18.89
		Upper	0.17149	0.14686	14.36
e97	5093 N	Near	0.22290	0.19081	14.39
		Upper	0.20767	0.18505	10.89

Across 1000, 850 and 500 hPa, u gains are 40.5%, 20.1% and approximately zero; v gives 40.7%, 19.6% and approximately zero. From 850 to 500 hPa, t falls from 12.3% to 0.1%, r from 15.1% to 7.4%: variable identity alone does not explain the split.

Classical controls and architecture. Near/upper G ratios are 9.14 for CRA5, 1.72 for matched MPCA-only, and 1.32–1.35 for non-matched MPCA+PCA (Table 2). The separate matched-rate MPCA+PCA point in Fig. 1 is omitted from this group table: its overall correction is only 3.7%, while MPCA-only is the main matched control. Distinct checkpoint weights share an architecture and residual objective. This supports a compressor-dependent pattern for this corrector, but its local receptive field may preferentially repair surface structure while missing broader upper-air errors. The MPCA architecture sweep does not isolate CRA5 receptive-field effects; we therefore test a fixed parameter-matched intervention below.

Parameter-matched receptive-field ablation. DilatedUNet2 changes only the two bottleneck convolutions to dilation/padding 2 and 3, preserving all other operations, shapes and 1,646,548 parameters. The same CRA5 residuals and training recipe select epoch 125 on 2017, stopping at 137 before testing. This fixed ablation is not architecture search. Training/validation takes 2,278 s. Both models use 19.86 G counted FLOPs/frame; same-run batch-8 inference averages 1.316/1.315 ms/frame.

Including pooling, bilinear resizing and skips, central theoretical support grows $72 \times 72 \to 96 \times 96$ cells (unclipped maximum spans $76 \to 100$ per axis).

Gradient diagnostics of $f(x)$ use eight fixed 2017 inputs, all 20 outputs and three positions: 480 probes/model. Squared gradients are summed over input channels and each probe normalised. Central mean 90%-energy Chebyshev radius decreases $3.41 \to 2.78$ cells. Across all probes, mean Dilated energy outside the original support is only 0.0033%. Expanded theoretical support does not imply broad effective use.

Overall correction is slightly worse (Table 1). In 2018/2022, upper G rises 0.30/0.31 percentage points (paired 95% intervals [0.27,0.33]/[0.28,0.34]), while near G falls 0.57/0.58 points. Near/upper ratios remain 8.28/8.24; differences are 27.79/28.30 points. $z500$ worsens ($G = -4.46\%/-5.12\%$), and upper $\bar{G}$ falls to 0.94%/1.01%. This ablation retains the split, but concentrated gradients and one seed leave broader architecture effects open.

Low-capacity affine control. One 3×3 convolution ($20 \to 20$, bias, no activation) has 3,620 parameters: *not parameter-matched*. The same recipe selects epoch 125 on 2017, stopping at 137; frozen direct19 gains are only 1.29%/1.31% in 2018/2022. Its 2018 upper $\bar{G} = -0.91\%$. Different capacity and context prevent attributing the U-Net gap solely to nonlinearity.

Spatial diagnostic. For a 5×5 box smoother S_5, the high-pass variance ratio $\mathrm{Var}(e - S_5(e))/\mathrm{Var}(e)$ is 57.6%/27.6% near/upper for CRA5, versus 22.2%/5.5% for unquantised e995. Since S_5 is not an orthogonal projection, this is not an additive variance partition or a causal mechanism.

7 Testing the Terrain Hypothesis

Larger reconstruction error need not imply a larger correctable fraction. We compare both against actual-orography roughness and retain the original proxy-elevation comparison as a sensitivity check.

Actual orography and roughness. ERA5 surface geopotential and land–sea mask are extracted from ARCO-ERA5 at 2018-01-01 00 UTC, with coordinates checked against the original 97×241 grid. Elevation is geopotential divided by $9.80665\,\mathrm{m\,s^{-2}}$; land is lsm > 0.5. Roughness is population elevation standard deviation in a 5×5 land-only neighbourhood with at least 13 valid neighbours. Ocean and out-of-domain cells are excluded without padding. Excluding 102 land cells leaves 15,343 cells overall and 7,573 west of 100°W.

Ten terrain-only quantile bins per region fix identical cells before and after correction. Per-channel normalised squared error is pooled over bin cells and all 1,460 test frames, square-rooted, then averaged over the 19 direct channels. The frozen checkpoint and 2014–2016 normalisation are unchanged.

Figure 2 shows larger baseline errors on rougher ground, except for a non-monotone low-roughness segment on all land. Between endpoint all-land deciles, baseline nRMSE rises from 0.08170 to 0.11482 and corrected nRMSE from

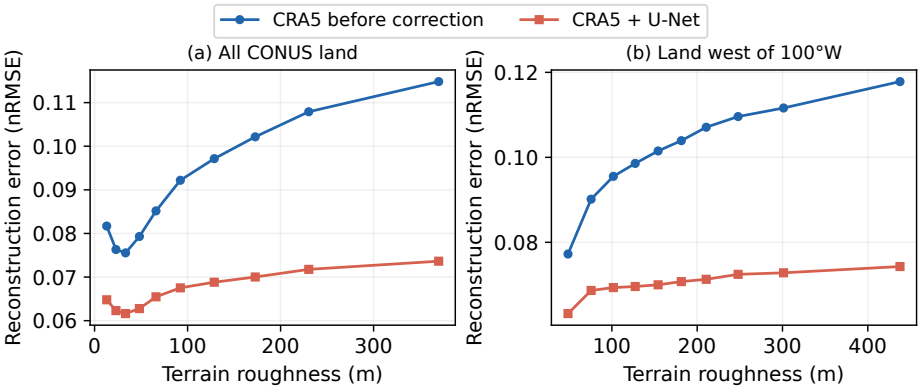


Fig. 2. 2018 frozen U-Net nRMSE before and after correction (19 direct channels). Roughness is 5×5 land-only elevation SD with at least 13 valid neighbours. Points show ten region-specific quantile bins at mean roughness; curves use identical cells and times within each panel. On western land, G rises from 18.1% to 36.9% between endpoint deciles. Restricting the region reduces east–west composition differences but does not establish causality.

0.06479 to 0.07364; G rises from 20.7% to 35.9%. On western land, baseline nRMSE rises from 0.07724 to 0.11783. Corrected error rises from 0.06323 to 0.07430; G increases from 18.1% to 36.9%. Correction narrows but does not remove the error gradient. All-land near-surface gain rises from 26.1% to 43.6%; upper-air gains remain 3.1–3.7% across deciles.

Elevation controls. Roughness/elevation Spearman correlation is 0.694 overall and 0.265 on western land. Western comparisons within fixed elevation bands (0–200, 200–600, 600–1200, 1200–2000, and above 2000 m) retain the broad association in several bands. For example, G rises from 14.3% to 36.8% across endpoint deciles at 600–1200 m. The pattern is not uniform: 0–200 m gains fluctuate around 30–35%, with only 29–31 cells per bin. These are descriptive controls, not independent spatial replications.

Legacy proxy-elevation sensitivity. Table 3 retains the original comparison at fixed western proxy-land positions.

The pressure-derived standard-atmosphere elevation and 1.0 K diurnal 2 m-temperature land threshold differ from the actual static fields above. The baseline surface-pressure high/low-elevation error ratio is 3.23× over CONUS, 2.28× on proxy-land, and 1.14× on western proxy-land; MPCA+PCA e995 gives 2.09, 1.70, and 0.95. This sensitivity check shows how the original proxy-based contrast changes with geographic restriction; it neither identifies a causal elevation effect nor replaces the actual-orography analysis.

Table 3. Legacy proxy-elevation control, western proxy-land, 2018: identical ground columns for near13/upper6, excluding `tcwv`. The pooled gain is $G_{\mathrm{pool}} = 1 - \sqrt{\langle a^2 \rangle / \langle b^2 \rangle}$, where a, b are per-cell normalised temporal RMSE after/before correction and $\langle \cdot \rangle$ pools channel–cell pairs. Unlike G, aggregation precedes the square root. Ratios use unrounded gains.

Elevation band	Cells	Near	Upper	Ratio
0–200 m	415	39.9%	5.0%	7.9×
200–600 m	850	39.2%	5.2%	7.6×
600–1200 m	2,311	33.5%	4.7%	7.1×
1200–2000 m	3,113	40.1%	4.7%	8.6×
above 2000 m	984	**41.9%**	**4.3%**	**9.7**×

Above 2000 m, eight low-level channels may include below-ground ERA5 extrapolation, potentially easing prediction. Proxy-sea gains also differ, 23.9% versus 5.5%: ground location alone does not explain the split.

8 Limitations

Published full-resolution CRA5 trains on 1979–2021 and tests on 2022–2023 [5]. Thus 2018 is held out for our corrector but overlaps CRA5 training. Dates are publication-reported, not verified from a checkpoint training log. Frozen full-year 2022 tests one unseen year, not robustness across climate regimes. One region, neural codec and few corrector architectures limit generalisation.

Mean-rate matching compares global/regional tasks, not identical domains. Per-frame rate coefficients of variation are 3.70% for MPCA-only and 0.95% for CRA5. Payload excludes fixed bases and correction weights; correction adds inference cost. Derived `tcwv` partly measures derivation error. No pointwise or per-block improvement is guaranteed.

Bootstrap intervals condition on fitted checkpoints and year; seven-day blocks do not establish independence beyond that lag. Spatial cells are correlated. Roughness covaries with elevation/geography; western lowland bins contain only 29–31 cells. Below-ground values complicate mountain comparisons. Both terrain analyses describe associations, not causality. Matched dilation expands theoretical support but retains concentrated gradients; one seed and dilation pattern cannot exclude broader context or optimisation effects. Unequal group sizes and error magnitudes affect the shared residual-MSE loss.

9 Conclusion

A frozen U-Net reduces direct19 CRA5 nRMSE by 25.8% in 2018 and 26.4% in unseen 2022, without extra frame bits or fine-tuning. Near/upper gains are 32–33%/3.5–3.6%. Matched bottleneck dilation retains a roughly eightfold split, concentrated gradient energy and slightly worse overall error. On western land,

ERA5-orography roughness is associated with larger baseline errors and correction gains; corrected errors remain higher on rough ground. Elevation-stratified checks do not establish a causal mechanism. Model-relative predictability persists in one year outside CRA5's reported training period; terrain causality, architecture and broader robustness remain open. Shared weights and computation are additional costs.

Code availability. Code and reproduction instructions are hosted at https://github.com/tj567356/CRA5-Decoder-Correction.

Disclosure of Interests. The authors have no competing interests.

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